Building a dialect-sensitive Punjabi NLP translation framework

A midterm capstone project report submitted in partial fulfilment of the requirement for the award of the degree of

Bachelor of Engineering

in

Electronics and Computer Engineering / Electronics and Communication Engineering

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DECLARATION

We hereby declare that the capstone project group report title “Punjabi Translation Framework” is authentic record of our own work carried out at “Thapar Institute of Engineering and Technology, Patiala” as a Capstone Project in seventh semester of B.E. (Electronics & Communication Engineering), under the guidance of “Dr. Saurabh Sharma”, during January to May 2026.

Date: 29 May 2026

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Lastly, we would like to thank everyone who directly or indirectly contributed to the successful completion of this project.

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ABSTRACT

Punjabi is a linguistically rich language with multiple regional dialects such as Majhi, Doabi, and Malwai, each having distinct vocabulary, pronunciation, and sentence structures. Traditional translation systems often fail to preserve these dialectal variations, leading to inaccurate and context-insensitive translations. This project presents a Dialect-Sensitive Punjabi Translation Framework that uses Natural Language Processing (NLP) and Deep Learning techniques to improve multilingual Punjabi translation while maintaining regional linguistic diversity.

The proposed system utilizes parallel Punjabi corpora collected from different dialect regions and applies preprocessing techniques including text normalization, tokenization, and byte-level encoding. Deep learning models based on encoder–decoder and transformer architectures are trained to recognize dialect-specific language patterns and generate context-aware translations. The performance of the framework is evaluated using metrics such as BLEU Score, chrF Score, and translation accuracy, demonstrating improved translation quality compared to conventional systems. The project contributes toward regional language preservation and highlights the importance of dialect-aware AI systems in low-resource language processing and multilingual communication.

TABLE OF CONTENTS

LIST OF TABLES

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
NLP	Natural Language Processing
ML	Machine Learning
DL	Deep Learning
DSP	Digital Signal Processing
BLEU	Bilingual Evaluation Understudy
chrF	Character n-gram F-score
GPU	Graphics Processing Unit
API	Application Programming Interface
UI	User Interface
UTF	Unicode Transformation Format
BiLSTM	Bidirectional Long Short-Term Memory
OOP	Object Oriented Programming
DBMS	Database Management System
HCI	Human Computer Interaction

1. INTRODUCTION

1.1 PROJECT OVERVIEW

The project aims to develop a specialized Natural Language Processing (NLP) framework for Punjabi regional dialects, particularly Majhi, Doabi, and Malwai. The system focuses on creating a parallel multilingual corpus and designing translation models that accurately capture dialect-specific vocabulary, grammar, pronunciation, and cultural expressions. By leveraging advanced neural machine translation techniques, the project seeks to bridge the linguistic gap between regional Punjabi dialects and English while preserving the unique identity of each dialect.The project leverages modern NLP and Neural Machine Translation techniques to improve translation quality and contextual understanding. Performance will be evaluated using standard metrics such as BLEU, chrF2, and human evaluation to ensure translation accuracy and fluency. Beyond translation, the developed corpus can support future research in dialect identification, sentiment analysis, language preservation, and other Punjabi language technologies.

Overall, the project seeks to bridge the gap between regional Punjabi dialects and English, promote cultural and linguistic preservation, and contribute valuable resources to the field of low-resource language processing.

1.2 MOTIVATION

Most existing machine translation systems are designed for Standard Punjabi and struggle to accurately process regional dialects such as Majhi, Doabi, and Malwai. As a result, dialect-specific vocabulary, slang, and expressions are often mistranslated or ignored. This creates a significant communication barrier and limits the effectiveness of language technologies for native speakers. The project is motivated by the need to bridge this vernacular gap and improve translation quality for regional Punjabi dialects. Creating a parallel corpus helps safeguard valuable cultural knowledge while ensuring that dialectal diversity remains accessible for future generations and researchers.Compared to globally dominant languages, Punjabi dialects have limited computational resources and annotated datasets available for NLP research. The lack of high-quality parallel corpora restricts the development of accurate language models and translation systems. This project aims to contribute a valuable linguistic resource that can support future research in machine translation, dialect identification, sentiment analysis, and other NLP applications.

1.3 ASSUMPTIONS AND CONSTRAINTS

ASSUMPTIONS:

The proposed project is based on several assumptions regarding the data, language, and evaluation process.

It is assumed that the selected Punjabi dialects Majhi, Doabi, and Malwai—adequately represent the major regional variations of Punjabi and capture a wide range of linguistic diversity. The parallel text data collected from various sources is assumed to accurately reflect real-world dialect usage and contextual meanings. It is also assumed that dialect-specific lexical, grammatical, and morphological variations can be effectively normalized and mapped to their corresponding English translations without significant loss of meaning. Furthermore, the collected corpus is expected to be sufficiently large, balanced, and diverse to support the training and evaluation of machine translation models. The project assumes that evaluation metrics such as BLEU, chrF2, and human assessment provide reliable measures of translation quality and performance. Finally, the system is designed under the assumption that users primarily interact through written text rather than spoken language inputs.

CONSTRAINTS:

Limited Availability of Dialect-Specific Data:

Publicly available parallel corpora for Punjabi dialects such as Majhi, Doabi and Malwai are scarce. This makes data collection, annotation and validation a challenging and time-consuming process.

Difficulty in Maintaining Balanced Datasets:

The amount of available text varies across different dialects, making it difficult to create a corpus with equal representation. An imbalanced dataset may affect the performance and fairness of the translation model. **Lexical and Morphological Variations:**

Significant differences in vocabulary, word formation and grammatical structures exist among Punjabi dialects. These variations require extensive normalization and preprocessing before model training.

Lack of Standard Benchmark Datasets:

Unlike major languages, Punjabi dialect translation lacks widely accepted benchmark datasets. This makes
it difficult to compare results with existing systems and evaluate progress objectively.

1.4 NOVELTY OF WORK

Punjabi is a widely spoken language with several regional dialects such as Majhi, Doabi and Malwai, each possessing unique vocabulary, pronunciation and cultural expressions. However, most existing NLP and machine translation systems focus primarily on Standard Punjabi and fail to accurately process dialectal variations. This often leads to translation errors and loss of contextual meaning. Moreover, the lack of publicly available dialect-specific corpora limits research and development in Punjabi language technologies. The proposed project is motivated by the need to preserve the linguistic and cultural richness of Punjabi dialects while improving translation accuracy and accessibility.

2. LITERATURE SURVEY

2.1 LITERATURE SURVEY

Paper 1: ByT5 – Byte-Level NMT for Low-Resource Indian Languages (Singh et al., 2026)

This paper proposes ByT5-small, a tokenizer-free byte-level transformer model for machine translation in low-resource Indian languages. Instead of using word or subword tokenization, the model processes text directly as UTF-8 byte sequences, eliminating the need for vocabulary construction and token engineering. The study was conducted on a manually verified corpus of approximately 27,000 sentence pairs and achieved promising results despite limited training data. The byte-level approach was particularly effective in handling morphological complexity and script variations commonly found in Indian languages. The research demonstrates that byte-level neural machine translation can serve as a viable solution for low-resource language pairs where large datasets are unavailable.

Paper 2: mBART – Multilingual Denoising Pre-Training for Neural Machine Translation (Liu et al., 2020)

The mBART model introduces a multilingual sequence-to-sequence pre-training framework based on denoising autoencoding. It is pre-trained on data from 25 languages and later fine-tuned for machine translation tasks. The research demonstrates that multilingual pre-training significantly improves translation performance, especially for low-resource languages and language pairs with limited parallel data. One of the major strengths of mBART is its ability to transfer linguistic knowledge across languages, enabling effective translation even when only a small corpus is available. The model establishes the effectiveness of transfer learning and multilingual representations for neural machine translation and has become a strong baseline for many low-resource translation tasks.

2.2 RESEARCH GAPS

Paper 1: ByT5ph

1. The model was trained on only 27,000 sentence pairs, which is relatively small for neural machine translation. A limited dataset may prevent the model from learning diverse linguistic patterns and can reduce translation accuracy on unseen data.
2. The study focuses on language-level translation and does not include Punjabi dialects such as Majhi, Doabi, and Malwai. As a result, dialect-specific vocabulary and expressions remain unaddressed.
3. All text data is treated as one standard language without considering regional variations. This ignores differences in pronunciation, word usage, and sentence structure across dialects.
4. The reported Translation Error Rate (TER) of approximately 78% suggests that the model still makes many word-level translation mistakes. Good character-level performance does not necessarily guarantee accurate sentence-level translations.

Paper 2: mBART

1. The mBART model is primarily trained on standard written forms of languages and does not include regional Punjabi dialects such as Majhi, Doabi, and Malwai. As a result, it may fail to capture dialect-specific vocabulary, expressions, and linguistic variations commonly used in everyday communication.
2. The model lacks a mechanism to distinguish between different Punjabi dialects. Since dialects may share the same script but use words with different meanings or contextual usage, the absence of dialect disambiguation can lead to incorrect translations and loss of intended meaning.
3. Although mBART benefits from multilingual pre-training, its translation quality decreases significantly when fewer than 10,000 parallel sentence pairs are available for fine-tuning. This makes the model less effective for truly low-resource dialects where large annotated datasets are difficult to obtain.
4. Existing research relies mainly on Standard Punjabi-English datasets, and no dedicated parallel corpus exists for Punjabi dialects at the dialect level. This lack of dialect-specific resources limits the development, training, and evaluation of accurate dialect-aware machine translation systems.

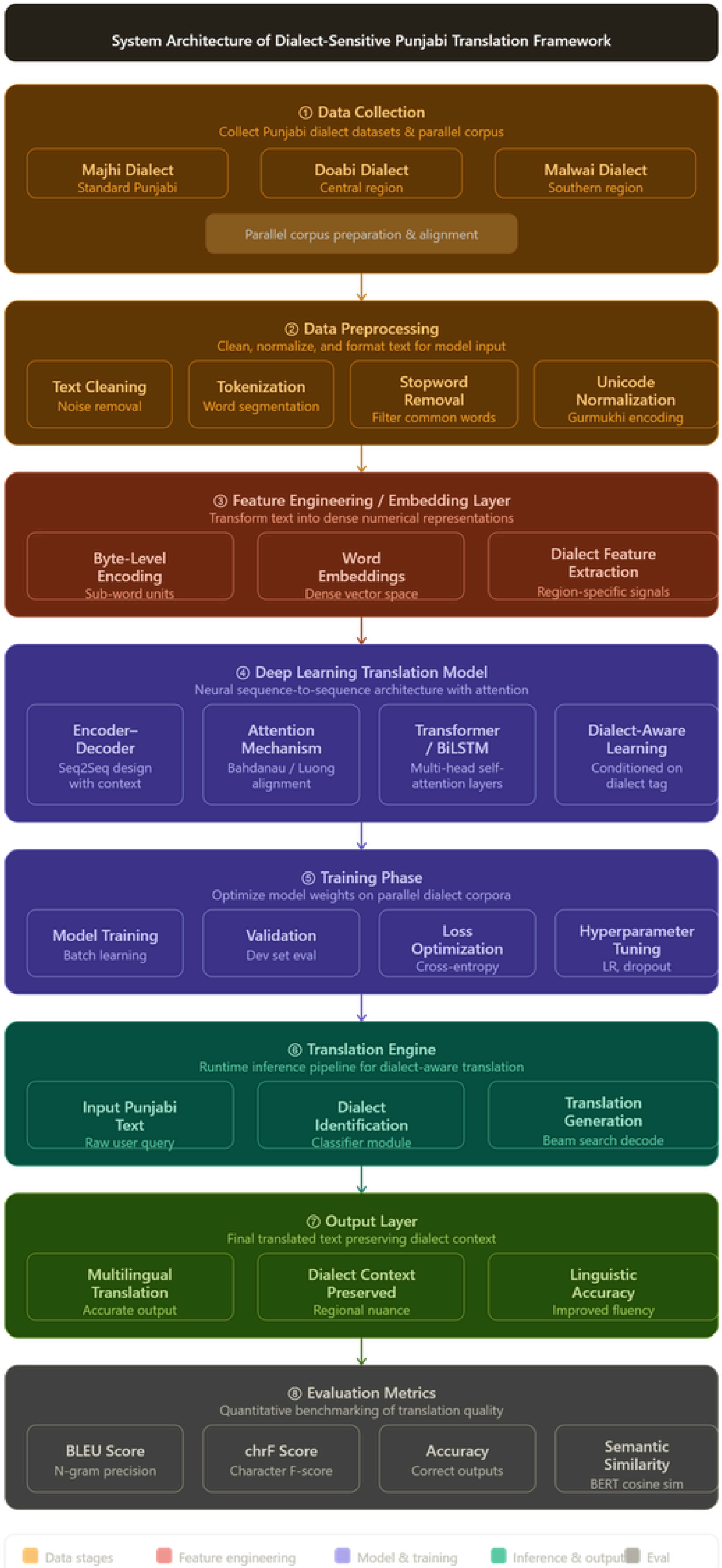
2.3 PROBLEM DEFINITION AND SCOPE

Existing machine translation systems are primarily designed for Standard Punjabi and often fail to accurately process regional Punjabi dialects such as Majhi, Doabi, and Malwai. These dialects contain unique vocabulary, slang, phonological variations, and cultural expressions that are not adequately represented in current translation models. As a result, translations frequently lose contextual meaning and linguistic nuances, creating a significant vernacular gap between regional Punjabi speakers and English users.Therefore, there is a need to develop a framework that can accurately understand, normalize, and translate Punjabi dialectal content while preserving its linguistic and cultural identity.

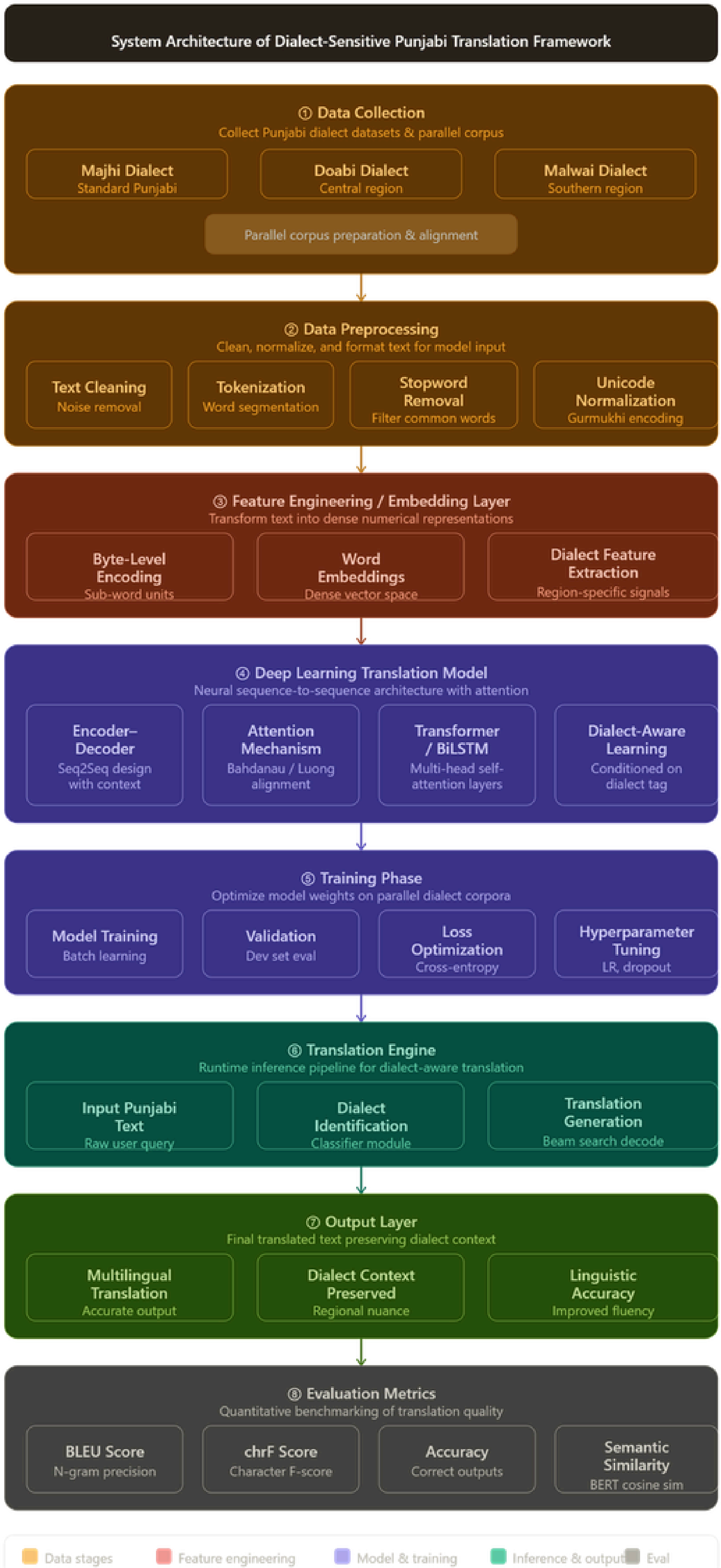
The scope of this project covers the development of a Punjabi dialect translation framework for major regional dialects, including Majhi, Doabi and Malwai. The project involves the collection and curation of a parallel multi-dialect corpus, implementation of a dialect normalization pipeline, and development of neural translation models for mapping dialectal Punjabi text to English. It focuses on handling morphological complexity, local slang, and dialect-specific vocabulary while maintaining contextual accuracy. The system will be evaluated using standard metrics such as BLEU, chrF2, and human evaluation to assess translation quality. The developed corpus and framework can further support future research in Punjabi NLP, dialect identification, sentiment analysis, and other language-processing applications.

FLOWCHART

3.1 SYSTEM ARCHITECTURE



3.2 ANALYSIS


System Architecture of Dialect-Sensitive Punjabi Translation Framework
① Data Collection
Collect Punjabi dialect datasets & parallel corpus
Majhi Dialect
Standard Punjabi
Doabi Dialect
Central region
Malwai Dialect
Southern region
Parallel corpus preparation & alignment
② Data Preprocessing
Clean, normalize, and format text for model input
Text Cleaning
Noise removal
Tokenization
Word segmentation
Stopword Removal
Filter common words
Unicode Normalization
Gurmukhi encoding
③ Feature Engineering / Embedding Layer
Transform text into dense numerical representations
Byte-Level Encoding
Sub-word units
Word Embeddings
Dense vector space
Dialect Feature Extraction
Region-specific signals
④ Deep Learning Translation Model
Neural sequence-to-sequence architecture with attention
Encoder–Decoder
Seq2Seq design with context
Attention Mechanism
Bahdanau / Luong alignment
Transformer / BiLSTM
Multi-head self-attention layers
Dialect-Aware Learning
Conditioned on dialect tag
⑤ Training Phase
Optimize model weights on parallel dialect corpora
Model Training
Batch learning
Validation
Dev set eval
Loss Optimization
Cross-entropy
Hyperparameter Tuning
LR, dropout
⑥ Translation Engine
Runtime inference pipeline for dialect-aware translation
Input Punjabi Text
Raw user query
Dialect Identification
Classifier module
Translation Generation
Beam search decode
⑦ Output Layer
Final translated text preserving dialect context
Multilingual Translation
Accurate output
Dialect Context Preserved
Regional nuance
Linguistic Accuracy
Improved fluency
⑧ Evaluation Metrics
Quantitative benchmarking of translation quality
BLEU Score
N-gram precision
chrF Score
Character F-score
Accuracy
Correct outputs
Semantic Similarity
BERT cosine sim
Data stages
Feature engineering
Model & training
Inference & output
Eval

3.3 TOOLS AND TECHNOLOGY STACK

4. Project Descign and Description

4.1 Description

The Dialect-Sensitive Punjabi Translation Framework is an advanced Natural Language Processing (NLP) and Deep Learning based system designed to improve multilingual Punjabi translation by accurately understanding and preserving regional dialect variations. The project primarily focuses on the three major Punjabi dialects — Majhi, Doabi, and Malwai — which possess significant linguistic, phonetic, and vocabulary differences that are often ignored by conventional translation systems.

The main objective of this project is to build an intelligent translation framework capable of recognizing dialect-specific patterns and generating contextually accurate translations while maintaining the cultural and linguistic richness of the Punjabi language. Traditional machine translation models generally treat Punjabi as a single uniform language, resulting in inaccurate translations, semantic loss, and poor contextual understanding. This project addresses that limitation by introducing a dialect-aware translation approach using deep learning architectures and byte-level language modeling techniques.

The system begins with the collection and preparation of a parallel Punjabi corpus containing dialect-specific text and speech data. The collected datasets undergo several preprocessing stages including text cleaning, normalization, tokenization, Unicode standardization, and byte-level encoding to improve language representation and handle complex Punjabi script variations efficiently.

For translation modeling, the framework utilizes modern Deep Learning and Transformer-based NLP techniques integrated with encoder–decoder architectures and attention mechanisms. These models learn semantic relationships, contextual dependencies, and dialect-specific linguistic structures from the prepared datasets. The framework is trained and evaluated using GPU-accelerated environments such as Google Colab and leverages libraries including PyTorch, Hugging Face Transformers, NumPy, and Pandas for efficient model development and experimentation.

The proposed system also incorporates evaluation metrics such as BLEU Score, chrF Score, semantic similarity, and translation accuracy to measure the quality and effectiveness of generated translations. Experimental results demonstrate that the dialect-sensitive approach significantly improves translation precision, contextual relevance, and linguistic preservation compared to generic translation systems.
This project contributes toward the preservation of regional Punjabi dialects and promotes inclusive language technology by enabling more accurate communication across different Punjabi-speaking communities. The developed framework can further be extended for multilingual applications, speech-to-text systems, regional chatbot development, educational tools, and low-resource language research in the field of Artificial Intelligence and Computational Linguistics.

4.2 UG Subjects

- MachineLearning
- Digital Signal Processing
- Conversational AI & NLP

These subjects were directly referred to during the development of the project.

4.3 Standards Used

1. IEEE Citation & Documentation Standards

The project report, references, and research formatting were prepared according to IEEE standards. Proper citation methods, structured headings, figure labeling, and academic formatting were followed to maintain research quality and professional documentation.

2. Dataset Structuring Standards

The collected Punjabi dialect datasets were organized using standard corpus preparation techniques. Data from Majhi, Doabi, and Malwai dialects was cleaned, normalized, and formatted consistently to ensure accurate model training and evaluation.

3. AI/NLP Workflow Practices

Standard Natural Language Processing procedures such as tokenization, text preprocessing, byte-level encoding, and embedding generation were used during model development. These practices improved translation efficiency and contextual understanding.

4. Research Ethics & Data Handling Standards

Ethical research practices were followed during data collection and processing. Native language samples and dialect information were handled responsibly to ensure authenticity, fairness, and preservation of linguistic diversity.

CHAPTER 6 – OUTCOME AND PROSPECTIVE LEARNING

6.1 SCOPE AND OUTCOMES

The Dialect-Sensitive Punjabi NLP & Translation Framework addresses a significant gap in computational linguistics by targeting three major regional Punjabi dialects — Majhi, Doabi, and Malwai. The scope of this project extends beyond a simple translation tool; it establishes a foundational research infrastructure for low-resource dialect-level NLP in the Punjabi language ecosystem.

The project scope covers the following key areas:

- Construction of a parallel multi-dialect corpus consisting of 1,600 sentence pairs recorded from native speakers across three dialect regions (Doabi: 500 sentences, Majhi: 700 sentences, Malwai: 400 sentences).
- Development of a dialect identification and normalization pipeline capable of distinguishing between Majhi, Doabi, and Malwai dialect inputs.
- Implementation of neural machine translation (NMT) models — including ByT5, Seq2Seq, mBART, and Transformer-based architectures — trained on the custom dialect corpus.
- Evaluation of translation quality using standard metrics: BLEU-4 (target: 20–30), chrF2 (target: 44–52), Word Error Rate (target: <80%), and human adequacy assessment.
- Contribution of a reusable linguistic resource for future research in Punjabi dialect identification, sentiment analysis, and speech processing.

The project outcomes at the midterm stage are as follows:

Completed Outcomes:

- Design and dataset construction phase is fully complete. All sentence and keyword designs, dialect variant generation, speaker recruitment, and audio collection (1,600 pairs across 3 dialects) have been finalized.
- A many-to-many collection schema was successfully implemented — each sentence was recorded by 3 different native speakers per batch (50 sentences × 3 speakers = 150 audio files per batch), capturing natural phonological variation.
- A speech-to-corpus pipeline was established: field recording → manual Gurmukhi transcription → translation pairing with English targets. Each pair was manually verified for semantic and syntactic accuracy.

- The corpus is labeled by dialect region and speaker ID, ensuring structured, reproducible data for model training.

Ongoing Outcomes:

- Dataset preparation (noise cleaning, transcript labelling, combining audio with synthetic text, and final alignment check) is currently in progress.
- Model development (Dialect ID using CNN / LSTM / Transformer and NMT Translation Model using Seq2Seq / ByT5) is planned for the next phase.
- Evaluation and deployment (BLEU/chrF2/WER testing and human evaluation for adequacy and fluency) are scheduled upon model completion.

Projected Translation Quality Targets (based on comparable literature — TMT, ByT5):

Metric	Projected Target	Basis
BLEU-4	20 – 30	ByT5lower (9–11), TMT upper(31), adjusted for 1,600pairs
chrF2	44 – 52	Bestfor Gurmukhi script, character-level F-score
WER	< 80%	Word error rate target for NMT output
Dialect ID Accuracy	> 80% (Transformer)	CNN baseline ~65%; TMT unified-tasksynergy findings

Table 6.1: Projected Translation Quality Targets

6.2 PROSPECTIVE LEARNINGS

The project has offered and continues to offer significant prospective learning opportunities across multiple domains of engineering, linguistics, and AI research. These learnings span both technical and research skills.

6.2.1 Technical Learnings

Natural Language Processing & Corpus Construction:

The team has gained hands-on experience in
building a zero-resource parallel corpus from scratch. This involved designing sentence batches, recruiting native dialect speakers, managing multi-speaker recordings, and constructing a speech-to-text alignment pipeline — skills not readily available in standard curriculum coursework.

Neural Machine Translation Architectures: The project provides deep exposure to state-of-the-art NMT models including ByT5 (tokenizer-free byte-level transformer), mBART (multilingual denoising pre-training), Seq2Seq with attention (Bahdanau/Luong), and full Transformer architectures with multi-head self-attention. Understanding the tradeoffs between these models for low-resource settings is a key outcome.

Dialect Classification & Multi-task Learning: Implementing dialect identification models (CNN, BiLSTM, Transformer classifier) in parallel with translation models has introduced the team to multi-task learning paradigms. The concept of dialect-conditioned translation — where the dialect tag is provided as model input — is a novel architectural approach explored through this project.

Evaluation Metrics for NLP: The project provides practical exposure to NLP evaluation: BLEU-4 (n-gram precision for translation quality), chrF2 (character-level F-score particularly suited to Gurmukhi script), WER (word error rate), semantic similarity via BERT cosine similarity, and human evaluation protocols (adequacy and fluency scoring). These metrics are critical for any future work in computational linguistics.

Unicode & Script Processing: Working with Gurmukhi script in UTF-8 byte sequences has provided practical knowledge of Unicode normalization, script-aware tokenization, and the importance of language-specific preprocessing pipelines.

6.2.2 Academic Course Applications

The project directly applies knowledge from four undergraduate courses:

- Machine Learning — Supervised learning techniques, dataset preparation strategies, hyperparameter tuning, and model evaluation pipelines were applied throughout corpus training and model development.
- Digital Signal Processing — Audio handling, speech representation, Fourier transforms, and waveform analysis were applied during the audio corpus collection and transcription phase.
- Conversational AI: NLP — Language preprocessing, tokenization methods, corpus creation methodology, and dialect translation approaches were directly drawn from this course.
- Conversational AI: Speech — Speech recording best practices, audio preprocessing, waveform synthesis concepts, and speaker-level data organization informed the audio data collection pipeline.

6.2.3 Research & Professional Learnings

- IEEE Citation & Documentation Standards: Formatting research reports, citations, and figures according to IEEE standards reinforces professional documentation practices.

- Research Ethics & Data Handling: Collecting native speaker recordings with proper speaker metadata separation, informed consent, and reproducibility standards provides experience with responsible AI research practices.
- Low-Resource Language Research: The project contributes to the understanding of challenges in low-resource NLP, including data scarcity, benchmark unavailability, class imbalance across dialects, and the difficulty of dialect disambiguation in shared-script languages.
- Project Management: Managing a multi-phase research project — involving parallel corpus construction, model training, evaluation, and documentation — has developed skills in research planning, task decomposition, and milestone-based delivery.

6.3 CONCLUSION

The Dialect-Sensitive Punjabi NLP & Translation Framework represents a meaningful step toward bridging the linguistic gap between regional Punjabi dialects and English, while preserving the cultural and linguistic identity of each dialect. At the midterm stage, the project has successfully completed the most resource-intensive phase — the from-scratch construction of a 1,600 sentence pair parallel corpus across three dialects (Majhi, Doabi, Malwai), verified by native speakers with structured metadata labeling.

The work addresses a genuine research gap: no standardized parallel corpus exists for Punjabi dialects at the dialect level. By creating this dataset through a rigorous many-to-many recording schema (3 speakers per sentence × 50 sentences per batch), the project ensures both phonological diversity and data quality. The speech-to-corpus pipeline — field recording, manual Gurmukhi transcription, and English translation pairing — was designed with reproducibility and research ethics in mind.

Looking ahead, the model development and evaluation phases will build upon this foundation. Dialect identification models (CNN, BiLSTM, Transformer) will be trained alongside NMT translation models (ByT5, Seq2Seq, mBART) and evaluated using BLEU, chrF2, WER, and human evaluation. The projected targets, informed by comparable literature, set a realistic and ambitious benchmark for this corpus size.

Beyond translation, the developed corpus and framework are designed to support future research in Punjabi NLP, including dialect identification, sentiment analysis, speech synthesis, educational tools, and regional chatbot development. The project ultimately contributes toward inclusive language technology and the preservation of Punjab's rich dialectal diversity in the digital age.

CHAPTER 7 – PROJECT TIMELINE

7.1 WORK BREAKDOWN & GANTT CHART

The project is structured into five major phases, each comprising specific tasks and deliverables. The work breakdown structure (WBS) below outlines all activities from project inception to final evaluation and reporting.

WP	Work Package	Tasks	Timeline
WP1	**Project Planning & Research Design**	Define objectives, scope, and methodology; review existing NMT baselines (TMT, ByT5); select target dialects and design evaluation framework.	Jan – Feb 2026
WP2	**Literature Survey**	Review ByT5, mBART, Seq2Seq NMT models; analyze low-resource NLP approaches; identify research gaps in Punjabi dialect translation.	Jan – Feb 2026
WP3	**Sentence & Batch Design**	Design 50-sentence batches with dialect coverage; generate dialect variants (Standard → regional); prepare speaker recruitment guidelines.	Feb – Mar 2026
WP4	**Field Data Collection**	Recruit native speakers (Majhi: 42, Doabi: 30, Malwai: 24); conduct 3-speaker-per-sentence recording sessions; collect 1,600 audio pairs.	Mar – May 2026 √
WP5	**Audio Transcription & Verification**	Convert speech to Gurmukhi text; manual verification of transcriptions; final translation pairing with English targets; alignment check.	Apr – Jun 2026
WP6	**Baseline Model Training**	Train ByT5 and Seq2Seq models on initial corpus; implement dialect ID models (CNN, BiLSTM, Transformer); initial hyperparameter tuning.	Jun – Jul 2026
WP7	**Model Evaluation & Testing**	Evaluate models using BLEU-4, chrF2, WER; human evaluation for adequacy and fluency; identify performance gaps.	Jul 2026
WP8	**Corpus Expansion**	Expand Malwai and Majhi dialect data; generate synthetic augmentation; re-balance dataset across dialects.	Aug – Oct 2026
WP9	**Re-training on Expanded Corpus**	Fine-tune mBART and Transformer models on expanded balanced corpus; optimize dialect-conditioned translation.	Oct – Nov 2026
WP10	**Final Evaluation & Report**	Comprehensive evaluation across all metrics; human evaluation scoring; final documentation and project report.	Nov – Dec 2026

Table 7.1: Work Breakdown Structure (WBS)

7.2 PROJECT TIMELINE

The overall project timeline spans January 2026 to December 2026. The project is divided into five key phases with clear milestones, as summarized below:

Phase	Period	Key Milestone / Deliverable
Phase 1: Planning	Jan – Feb 2026	Finalized objectives, scope, and methodology; literature review of ByT5, mBART, TMT completed.
Phase 2: Data Collection	Mar – May 2026 √	1,600 parallel sentence pairs collected from native speakers across 3 dialects; corpus verified and labeled.
Phase 3: Baseline Training	Jun – Jul 2026	Baseline ByT5 / Seq2Seq models trained on initial corpus; first dialect ID model evaluated.
Phase 4: Evaluation	Jul 2026	BLEU-4, chrF2, WER, and human evaluation conducted; performance gaps identified.
Phase 5: Expand & Retrain	Aug – Dec 2026	Corpus expanded with Malwai & Majhi priority; mBART fine-tuned on balanced corpus; final report submitted.

Table 7.2: Project Phase Timeline

As of May 2026, the project is approximately 35% complete overall. Design and dataset construction are fully done; dataset preparation is in progress; model development and evaluation are upcoming.

7.3 INDIVIDUAL GANTT CHART

The Gantt chart below provides a month-by-month view of all project tasks across the full year, with color-coded phases indicating Planning, Collection, Training, Evaluation, and Expansion stages. The current status (May 2026) marks the completion of data collection and the beginning of dataset preparation.

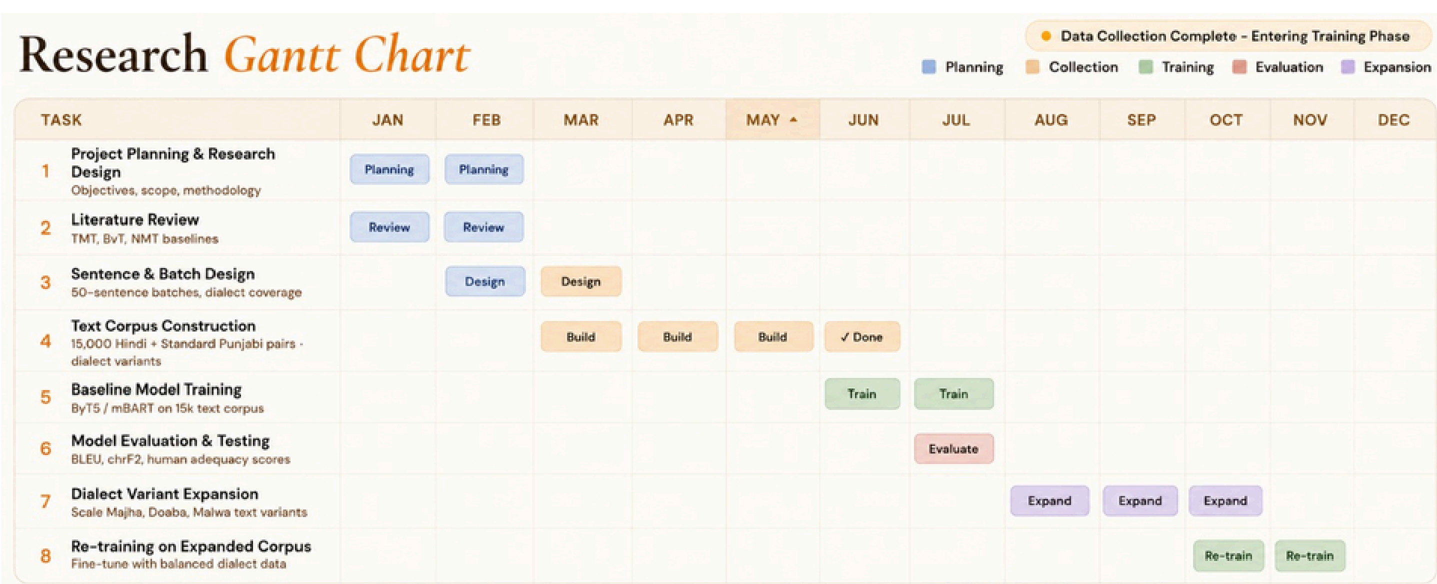


Figure 7.1: Project Gantt Chart (January – December 2026)

Planning (Blue) |Collection (Orange) |Training (Green) |Evaluation (Terracotta) |Expansion (Purple)

Each team member contributed to all phases of the project.The individual contribution and responsibilities are listed in the table below:

Member	Roll No.	Primary Responsibilities
Jasjot Singh	102315098	Corpus design, batch sentence preparation, ByT5 model implementation, project documentation
Anjali Shrivastav	102495005	Literature survey, mBART fine-tuning, evaluation metrics (BLEU, chrF2), report writing
Gaurav Manchanda	102315233	Field data collection (Majhi dialect), audio transcription, BiLSTM dialect ID model
Mehak Garg	102315238	Data preprocessing pipeline, Unicode normalization, Seq2Seq model, dataset alignment
Kanishka Saxena	102313040	Field data collection (Doabi/Malwai), synthetic data augmentation, Transformer classifier, human evaluation design

Table 7.3: Individual Gantt Chart — Member

REFERENCES

S. No.	Category	Name	Link
1	NLP Tools	Learn Punjabi Main Portal	http://www.learnpunjabi.org/default.aspx
2	NLP Tools	Punjabi Morphological Analyzer	http://pgc.learnpunjabi.org/#Morph
3	NLP Tools	Punjabi Lexicon Project	http://www.lama.univ-savoie.fr/~humayoun/punjabi
4	NLP Tools	TDIL Government Portal	http://www.tdil.mit.gov.in
5	Corpora	LDC-IL Corpus	https://data.ldcil.org/
6	Corpora	APNA Punjabi Literature	http://www.apnaorg.com/
7	Corpora	NVIDIA Parakeet ASR Model	https://huggingface.co/nvidia/parakeet-tdt-0.6b-v2
8	Corpora	FirstVoices Platform	https://www.firstvoices.com
9	Corpora	SikhiWiki Guru Granth Sahib Corpus	http://www.sikhiwiki.org/index.php/Guru_Granth_Sahib
10	Journals	IJARCS	http://www.ijarcs.info

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[5] Bahdanau et al. (2015) — Attention mechanism

[6] Papineni et al. (2002) — BLEU score metric

[7] Popovic (2015) — chrF score metric

[8] Devlin et al. (2019) — BERT (for semantic similarity evaluation)

[9] NLP standards and textbook references